

sTMS Interface Functional Specification

Singapore Polytechnic – Library System

Version 1.0

Document Version

Version	Effective Date	Summary of Changes	Created/ Updated By
0.1	14/07/2022	Initial version of SP library system IFS	AvePoint
0.2	11/09/2022	Drop field of 'Student Mobile'. Revise sample file. Change AccessNo from CAPES NO. to SP student virtual card NO. (numeric) for barcode scanner.	AvePoint
0.3	15/09/2022	Updated student enrollment status value list.	AvePoint
0.4	16/09/2022	Updated the filter condition of PT and NF files.	AvePoint
1.0	16/09/2022	Official version for acceptance and sign-off.	AvePoint

Document Acceptance Form

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Table of Contents

Document Version	1
Document Acceptance Form	2
1. Introduction	4
1.1 Purpose of this document	4
1.2 Intended Audience	4
2. Technical Overview	5
2.1 System Flow	5
2.2 Interface Design Summary	6
2.3 Business Rule	7
2.4 Interface File Generation Rule	8
3. Interface Fields	9
4. Sample Files	14

1. Introduction

1.1 Purpose of this document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirement and detailed technical specification between the SaaS shared Training Management System (sTMS) and Singapore Polytechnic Library System to interface students data for creation of Continuing Education & Training (CET) students' library accounts.

sTMS shall identify students' details based on the UAM Student ID, enrolment and graduation information and personal information, such as name and mobile number, to generate corresponding TXT file. sTMS shall generate and deposit the TXT file encrypted by SLIFT protocol in the SFTP server accordingly. Singapore Polytechnic can create student library account based on the transferred TXT file.

1.2 Intended Audience

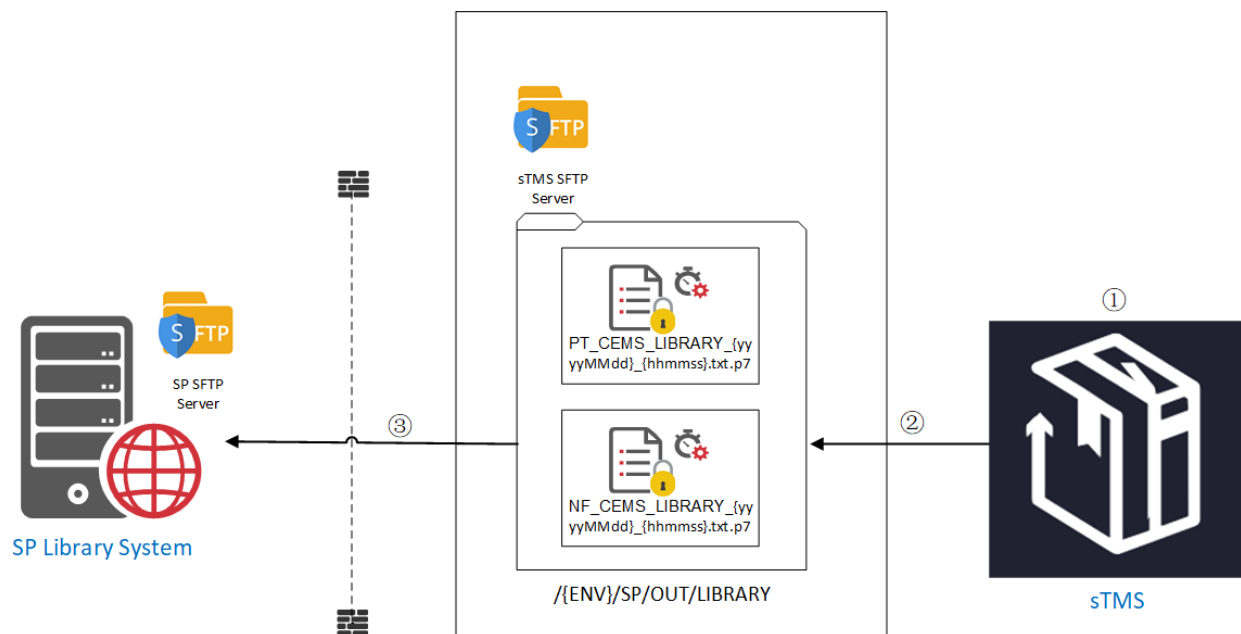
This technical document is used for system communication. Therefore, the intended audiences are the technical stakeholders of SP Library System and sTMS.

2. Technical Overview

2.1 System Flow

There are 3 roles involved in the interface diagram.

- **SP Library System:** Singapore Polytechnic Library system.
- **sTMS SFTP Server:** A SFTP server hosted in Azure that stores encrypted TXT file used for data exchange.
- **sTMS:** shared Training Management System.



SN	Description
1	sTMS generates the library TXT files based on student's enrollment status and course types on a pre-defined schedule.
2	sTMS configures timer job to push library files into sTMS SFTP server after encryption with SLIFT protocol.
3	sTMS SFTP server push library files into SP SFTP server.

2.2 Interface Design Summary

Output Interface File

Interface File Name	PT_CEMS_LIBRARY_{yyyyMMdd_hhmmss}.txt NF_CEMS_LIBRARY_{yyyyMMdd_hhmmss}.txt
Data Source System	shared Training Management System
Data Destination System	Singapore Polytechnic Library System
Mode of Interface	TXT file, with “ ” as delimiter
Method	SFTP
File Path	/ {ENV} / SP / OUT / LIBRARY ENV: UAT, PROD
Encryption Method	PGP
Data Syncing Mode	full load data sync
Frequency	Daily
Timing	9 pm
File Retention Policy	Files are retained for a maximum of 30 days in the sTMS SFTP server, after this period the files will be automatically deleted
File Transfer Policy	sTMS shall send an empty file with header info if no student related info should be sent to SP Library System
Possible Exception and Exception Handling	1. File loss or transfer failure due to sTMS internal exception. E.g. Timer service down, etc. Exception handling steps: • sTMS support team shall be informed by email of the transfer failure. • AP team checks the relevant logs to troubleshoot the cause of the failure. • If the issue is due to the transfer file could not be found in SP SFTP server: • sTMS support team will manually rerun the timer job to trigger file generation and deposit the file in SP SFTP server. 2. Data error. Exception handling steps: • sTMS support team shall be informed by email of the file generation failure. • AP team checks the relevant logs and file to troubleshoot the cause of the failure (data error types are listed below). • Field is mandatory but missing

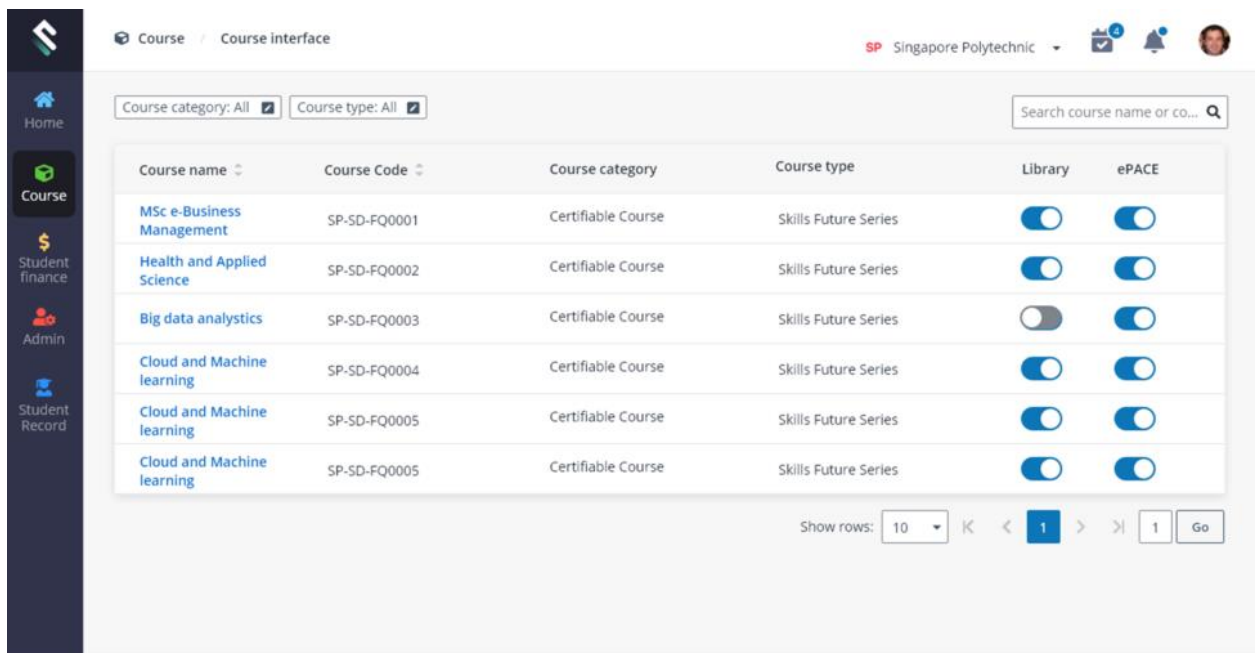
	- Field failed validation - AP redeposits the correct file in SFTP.
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2.3 Business Rule

- The student information would be sent if user toggles the course to be synced with Library interface as shown in the illustration below. By default, all set as Off.

Below will be default set as On

Course Category	Course Type
Part-time diploma	All
Specialist diploma	All
Advanced diploma	All
Diploma Conversion	All
Work-Study Programme	All



Course / Course interface

SP Singapore Polytechnic

Course category: All Course type: All

Search course name or co...

Course name	Course Code	Course category	Course type	Library	ePACE
MSc e-Business Management	SP-SD-FQ0001	Certifiable Course	Skills Future Series	☒	☒
Health and Applied Science	SP-SD-FQ0002	Certifiable Course	Skills Future Series	☒	☒
Big data analytics	SP-SD-FQ0003	Certifiable Course	Skills Future Series	☐	☒
Cloud and Machine learning	SP-SD-FQ0004	Certifiable Course	Skills Future Series	☒	☒
Cloud and Machine learning	SP-SD-FQ0005	Certifiable Course	Skills Future Series	☒	☒
Cloud and Machine learning	SP-SD-FQ0005	Certifiable Course	Skills Future Series	☒	☒

Show rows: 10 < 1 > 1 Go

2. sTMS will send the following two files containing students' information to SP Library system as follows:
 - a. Students attending the course from Part-time diploma (PT) Specialist diploma (SD), Advanced diploma (AD), Diploma conversion, and Work-study Programmes would be recorded in PT_CEMS_LIBRARY_{yyyyMMdd_hhmmss}.txt
 - b. Students attending the course from certifiable courses, non-certifiable courses would be recorded in NF_CEMS_LIBRARY_{yyyyMMdd_hhmmss}.txt

2.4 Interface File Generation Rule

Flow to Generate Library TXT File

SN	Description
1	sTMS identifies the student's enrollment status daily. Once the student is enrolled, sTMS shall locate the corresponding student information and course information. Refer to 3. Interface Fields for the fields to be sent.
2	sTMS generates a TXT file to consolidate the full-load data of students.
3	sTMS deposits the TXT file in SFTP server accordingly.

3. Interface Fields

SP Field	sTMS Name	Description	Data Type	Remarks
iChat Email	Email	UAM login ID	String(\$nvarchar(50))	If UAM have a valid official email address, interface shall send to this email to SP library for notification purpose.
Official Name	Name	Student's Full Name	String(\$nvarchar(50))	Sample: John Smith
Department Code	School	School Name	String(\$nvarchar(20))	Sample: EEE
Year of Study	Year of Entry	Year when student is admitted to the course	String(\$nvarchar(25))	Sample: 2022
Patron ID		Group value: Permanent/ Contract/ Student	String(\$nvarchar(25))	Leave as blank
Email ID/Ichat Email	Email	Student Personal Email	String(\$nvarchar(50))	If UAM doesn't have a valid email address, SP library shall use this personal email for notification purposes.
Course Code	CourseCode	Course Code	String(\$nvarchar(20))	Sample: SP-FQ0001
Modules	ModuleCode	Module Code	String(\$nvarchar(20))	Sample: BM001; BM002
Graduation Date	GraduationDate	Student's estimated graduation date	String(YYYY-DD-MM)	Sample: 2023-06-30
Student Status	EnrolmentStatus	Student enrolment Status Value list: CN – Enrolled GR – completed S – Suspended	String	For the student who has enrolled the current course, the interface will send 'CN'. After course completed, the value would be GR.

		WD - Withdrawn, Transferred		
SP access NO.	AccessNo	sTMS Student Virtual card NO.	String	Sample value: 1234567140922, 7 digits running number as access number & expired date in the form of ddmmyy The format to be confirmed, pending on the final solution of sTMS student barcode.
SPICE ID	StudentID	Student UAM ID	String(\$nvarchar(20))	Sample: U123456



My profile / Personal particulars




All courses

Dashb...

My course

Applic...

Requests & task

Transa...

Certific...

Calendar

Resour...

Personal particulars

My wallet

My letters

Announcements

Notifications

Personal particulars (Student ID: T123456)



Upload your photo

Photo modified date: 01/04/2022

Personal information

Manual request to update Sync from MyInfo

Name	Jonah.Louis	Date of birth	20/02/1990
Citizenship type	Foreigner	NRIC/FIN No.	12*****78
Pass type	PEP	Pass expiry date	01/01/2030
Nationality	China	Country of birth	Singapore
Race	Chinese	Gender	Male
Passport No.	E1*****89		

Supporting documents

NRIC.pdf

15.62KB

Supporting document. pdf

15.62KB

Other information

Edit

Contact info

Mobile number	+65 15454554	Email	Jonah.Louis@email.com
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Residential address

Course / Academic progress / Academic overview

Nanyang Polytechnic

Dashb...

Course

Student finance

Admin

Student record

Company record

Report

Course intake setup

Course intake management

Timetabling management

Application management

Academic progress

Enrolment management

Request management

Enrollment status: All

Search by student name...

BOE endorsed

Export

Publish

Student name	Student ID	Class	Enrollment status	Enrolment status reason	Endorsement status	Progression to next sem?
VC Valentin Connelly	A001	Class 1	Enrolled		Endorsed	N
DL Declan Lehner	A002	Class 1	Enrolled		Endorsed	
EG Eulah Glover	A003	Class 1	Withdrawn	Requested by student	Endorsed	
DF Devin Fisher	A004	Class 1	Enrolled		Endorsed	

Total items: 4

Show rows: 10

1

Go

Course / Course intake configuration

Home

Course

Student finance

Admin

Student record

Module configuration

Course configuration

Course intake configuration

Timetabling configuration

Catalogue configuration

Venue management

MER bank

Course category: All

Course type: All

Course start date: All

Status: All

Search course name or co...

Create

Import

Export

Edit

Copy

Assign role

Extend

Postpone

Generate URL

Course intake No.	Course name	Course code	Course category	Course type	Registration duration	Course duration	Assigned role	Status
TP-SD-FQ0001-220001	MSc e-Business Management	TP-SD-FQ0001	Certifiable Course	SkillsFuture Series	01/05/2021 - 08/05/2022	08/05/2021 - 10/08/2022		Draft
TP-SD-FQ0002-220001	Data Science and Digital Operation	TP-SD-FQ0002	Certifiable Course	SkillsFuture Series	01/05/2021 - 08/05/2022	08/05/2021 - 10/08/2022		Open for application
TP-SD-FQ0003-220001	Cloud and Machine learning	TP-SD-FQ0003	Certifiable Course	SkillsFuture Series	01/05/2021 - 08/05/2022	08/05/2021 - 10/08/2022		Confirmed
TP-SD-FQ0004-220001	Cloud and Machine learning	TP-SD-FQ0004	Certifiable Course	SkillsFuture Series	01/05/2021 - 08/05/2022	08/05/2021 - 10/08/2022		View supporting document
TP-SD-FQ0005-220001	Cloud and Machine learning	TP-SD-FQ0005	Certifiable Course	SkillsFuture Series	01/05/2021 - 08/05/2022	08/05/2021 - 10/08/2022		Cancelled
TP-SD-FQ0006-220001	Cloud and Machine learning	TP-SD-FQ0006	Certifiable Course	SkillsFuture Series	01/05/2021 - 08/05/2022	08/05/2021 - 10/08/2022		Application closed

Show rows: 10

1

Go

11

 Home

 Course

 Student finance

 Admin

 Student Record

Course / Course intake configuration / Create course intake

Course intake setup

Course intake management

1 General information

2 Course details

3 Q1 Registration information

4 Q1 Application information

5 Q2 Application information

6 Declaration form

7 Course intake fee

Deferred module return

Step 2. Course details

Course structure

Course certification

Pathway 1 (Normal course pathway)

Minimum requirement

By module number

Number of elective module

1

Class period setup

Class assignment setup

Semester setup

☐	Module name	Version	Semester	Module ID	Module school	Credit unit	Module type	Class period	
☒	e-Business Fundamentals	1	2022-1	BM0001	School of business	4	Core	Monday 18:00 -19:00	+2
☐	Digital Development	2	2022-1	BM0002	School of business	4	Elective	Monday 18:00 -19:00	+2

Auto create class once class exceeds 20 students

Pathway 2 (Normal course pathway)

Minimum requirement

By modular credit unit

Number of elective credit unit

0

Class period setup

Class assignment setup

Semester setup

☒	Module name	Version	Semester	Module ID	Module school	Credit unit	Module type	Class period
☒	e-Business Fundamentals	1	2022-1	BM0001	School of business	4	Core	Monday 18:00 -19:00

Class period

Monday 18:00 -19:00;

Friday 18:00 -19:00;

Saturday 09:00 - 10:00

Cancel

Save as draft

Back

Save and next

 Dashboard

 All courses

 Dashboard

 My course

 Academic advisement

 Application

 Requests & Task

 Transaction

 Academic document

 Award



Dashboard

My tasks 4

View all

Progression letter

Course: TP-SD-FQ0001

Created on 24/01/2022

Request returned

LOAS_20220302205

Course: TP-SD-FQ0001

Created on 24/01/2022

My recent applications



TP

Academic certificate

[TP-SD-FQ0001-220001]

Incomplete submission

Jonah.Louis

Student ID: S300053

RP

12/05/2022 12:00

1 2 0 0 - 3 4 5 4 - 0 2 3 0 - 1 2 3 3

Course: [TP-SD-FQ0001] Business Analytics for Management;
[TP-SD-FQ0001] Business Analytics for Management;
[TP-SD-FQ0001] Business Analytics for Management

Close

My virtual card

Republic Polytechnic



Enlarge

View profile

Announcement

See all

System maintenance ...

NYP

24/05/2022

>Lorem ipsum dolor sit amet,

consetetur sadipscing elitr, sed

di...nonumy eirmod tempor invidunt

New course semester... start!

NYP

24/05/2022

>Lorem ipsum dolor sit amet,

consetetur sadipscing elitr, sed di...

nonumy eirmod tempor invidunt ut

4. Sample Files



PT_CEMS_LIBRARY_20NF.CEMS_LIBRARY_20
221207_210959.txt 211207_210951.txt